Aditya Prasad

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U.S. Citizen

Education

06/2024 – Present **Ph.D. in Computer Science**, University of Chicago
Advisor: Haifeng Xu
Research Interests: Algorithmic Game Theory; Online and Stochastic Optimization; Forecasting and Evaluation

08/2020 – 05/2024 **B.S. in Computer Science**, University of Southern California
Magna Cum Laude; W.V.T. Rusch Engineering Honors Program
Undergraduate advisor: Shaddin Dughmi

Research

$(\alpha\beta)$ denotes alphabetical author order, as is standard in computer science theory.

5. **Prophet Arena-Pro: Scaling Towards A Comprehensive, Trustworthy, and Robust Ecosystem for Forecasting**
Anri Gu, Aditya Prasad, Qingchuan Yang, Haifeng Xu, Jibang Wu
In submission, 2026

4. **Batched Pandora's Box**
$(\alpha\beta)$ Shaddin Dughmi, Yusuf Hakan Kalayci, Vasilis Livanos, Aditya Prasad
In submission, 2026 · arXiv

3. **The Keychain Problem: On Minimizing the Opportunity Cost of Uncertainty**
$(\alpha\beta)$ Ramiro Deo-Campo Vuong, Robert Kleinberg, Aditya Prasad, Eric Xiao, Haifeng Xu
EC, 2026 · arXiv

2. **dpvis: A Visual and Interactive Learning Tool for Dynamic Programming**
David H. Lee, Aditya Prasad, Ramiro Deo-Campo Vuong, Tianyu Wang, Eric Han, David Kempe
SIGCSE TS, 2025 · arXiv

1. **On Supermodular Contracts and Dense Subgraphs**
$(\alpha\beta)$ Ramiro Deo-Campo Vuong, Shaddin Dughmi, Neel Patel, Aditya Prasad
SODA, 2024 · arXiv

Experience

06/2024 – Present **Graduate Research Assistant**, University of Chicago, Chicago, IL

- Designing robust and optimal contracts for Markov chain environments where the contract induces the transition kernel, given information only about the marginal state distribution (with Haifeng Xu and Krishnamurthy Iyer).
- Introduced the Keychain Problem and established connections to known theoretical questions on bipartite graphs. Proved hardness of variants and designed approximation algorithms (with Haifeng Xu and Robert Kleinberg).

06/2026 – 09/2026 **Research Intern**, Sooth Labs, Pittsburgh, PA

- Designed difficulty robust scoring systems for LLM forecasters, improving accuracy and speed of ranking forecasters and determining each one's individual quality.
- Developed heuristics to identify informative questions to improve ranking efficiency.
- Developed a scoring model for a live LLM forecasting benchmark, improving sample efficiency by 5.5× and halving pairwise ranking errors from 9% → 4%.

05/2022 – 05/2024 **Undergraduate Research Assistant**, University of Southern California, Los Angeles, CA

- Developed the state-of-the-art algorithm for obtaining the optimal contract for the supermodular single-agent contracts problem. Proved APX-hardness of the supermodular multi-agent contracts problem and developed an additive-PTAS for a special variant (with Shaddin Dughmi).
- Studied and developed new cutting-plane methods for linear programs to identify novel structural properties of optimal vertices. Wrote bachelor's thesis on applying these properties to strongly polynomial-time algorithms (with Vatsal Sharan).
- Designed and implemented a general-purpose dynamic programming visualization library, and used the library as a supplementary learning tool for a cohort of students in Algorithm Design (with David Kempe).

07/2019 – 05/2020 **Remote Research Assistant**, Purdue University, Princeton, NJ

- Completed a directed reading of *An Introduction to Statistical Learning* by Gareth James to learn foundational machine learning techniques. Combined machine learning algorithms in ensemble learners to evaluate efficacy in advertising (with Wreetabrata Kar).

Manuscripts

05/2024 **Contract Theory and Linear Programming: In Search of Strongly Polynomial-Time Algorithms**
Aditya Prasad, Shaddin Dughmi (Research Advisor), Sandeep Gupta (Honors Advisor)

05/2022 **An Overview of Low-Rank Matrix Completion using Convex Optimization Techniques**
($\alpha\beta$) Ramiro Deo-Campo Vuong, David H. Lee, Aditya Prasad, Yibo Wen

Teaching

03/2026 – 06/2026 **Math Foundations of ML** (CMSC 25300/35300) – Teaching Assistant
University of Chicago; Haifeng Xu and Michael Maire

01/2025 – 03/2025 **Learning, Decisions, and Limits** (DATA 37200) – Teaching Assistant
University of Chicago; Haifeng Xu and Frederick Koehler

06/2023 – 08/2023 **Discrete Mathematics** (CSCI 170) – Course Producer
University of Southern California; Shaddin Dughmi

01/2022 – 05/2023 **Algorithms** (CSCI 270) – Course Producer
University of Southern California; David Kempe and Shahriar Shamsian

Projects

2023 – 2024 **dpvis – A Dynamic Programming Visualizer**
dpvis.readthedocs.io
- Developed an open-source Python library featuring auto-generated interactive visualizations for arbitrary dynamic programs, including self-testing features that prompt users with interactive questions about the algorithm's execution.
- Designed a decoupled, event-driven architecture with **Plotly Dash + Python** supporting modular feature expansion and maintainability.
- Used industry-standard development practices: **pytest** for comprehensive unit testing and **mkdocs** for professional documentation.
- Released on PyPI in December 2023 for future use by students in USC's CSCI 270.

2022 **University Simple C Compiler**
- Built a custom compiler in C++ supporting SSA form, optimization passes, and register allocation.
- Implemented a recursive descent parser, symbol tables for semantic analysis, and LLVM intermediate representation generation.

2022 **Low-Rank Matrix Completion via Convex Optimization**
- Scraped NBA statistics using **Selenium** and **BeautifulSoup** to evaluate matrix completion on real-world environments.
- Applied convex optimization techniques (nuclear norm minimization) to predict NBA game outcomes using offensive rating and pace statistics.
- Extended methods to reconstruct 3D tensors for color voxel grids with up to 80% missing data, with RMSE < 37.

Talks

2026 **The Keychain Problem: On Minimizing the Opportunity Cost of Uncertainty**
Economics and Computation (EC) 2026

2025 **dpvis: A Visual and Interactive Learning Tool for Dynamic Programming**
SIGCSE TS

2024, 2025 **On Supermodular Contracts and Dense Subgraphs**
UChicago Theory Lunch (2025); USC Theory Lunch (2024)

2024 **In Search of Strongly Polynomial-Time Algorithms**
W.V.T. Rusch Engineering Honors Thesis Talk, USC

Honors

2023 – 2024 W.V.T. Rusch Engineering Honors Program

2022 NSF Summer Research Grant

2020 – 2024 USC Viterbi Dean's List

Graduate Level Coursework

University of Chicago
Approximation Algorithms, Mathematical Toolkit, Responsible Use of Data, Online Learning Operations and Electronic Markets, Advanced Algorithms, Advanced Operating Systems

University of Southern California
Convex and Combinatorial Optimization, Advanced Algorithms, Frontiers of Machine Learning, Theoretical Machine Learning

Service

Conference Reviewer: SODA (2027)
Journal Reviewer: JAAMAS (2026)
Workshop Reviewer:
Forecasting as a New Frontier of Intelligence Workshop *(at ICML 2026)*,
Economics of Modern ML: Markets, Incentives, and Generative AI *(at AAAI 2025)*

Skills

Languages: Python, C/C++, Java, R, Bash
Libraries: Plotly, Matplotlib, Dash, NumPy, TensorFlow, scikit-learn, pandas, Keras